

User Guide

JALoP over BEEP (v1.x)

26 January 2026

This Page is Intentionally Left Blank

Contents

1	JALoPv1.x.....	6
2	Download/Clone Source Code	6
3	Build and Install C Publisher (jald) & C Subscriber (jal_subscribe)	6
3.1	System Provisioning.....	6
3.1.1	Install Rust Dependencies	6
3.1.2	RHEL 7 / CentOS 7	7
3.1.3	RHEL 8 / CentOS 8	8
3.1.4	RHEL 9.....	9
3.2	Build and Install C Publisher & C Subscriber.....	10
3.3	Configure C Publisher	10
3.4	Configure JAL-Local-Store.....	10
3.5	Configure C Subscriber	11
4	Build and Install BeepCore Java	11
4.1	System Provisioning.....	11
4.2	Clone “beepcore-java” repository from GitHub.....	11
4.3	Install BeepCore Java from pre-built jar files	12
5	Build and Install Java Subscriber and Publisher (jnl_test-*.jar)	12
5.1	Clone “jjnl” repository from GitHub	12
5.2	Build and Install Java Subscriber (jnl_test)	13
5.3	Configure jnl_test-*.jar as a Subscriber.....	13
5.4	Configure jnl_test-*.jar as a Publisher.....	13
6	Run Publisher and Subscriber to Transfer JAL Records	14
6.1	Disable SELinux or set to Permissive (For Testing Only).....	14
6.2	Stop Firewall (For Testing Only)	14
6.3	Start JAL-LOCAL-STORE.....	14
6.4	Insert Records into JAL-Local-Store	14
6.4.1	Insert Log Records	14
6.4.2	Insert Audit Records.....	15
6.4.3	Insert Journal Records.....	15
6.5	Check for Inserted Records in JAL-Local-Store	15
6.6	Start Rust Inline Filter (Recommended)	15
6.7	Start C Publisher	16

6.8	Start Subscriber	16
6.9	Purge Records from Local Store	17
6.10	Run Java Publisher and Java Subscriber	17
6.10.1	Add Records to Java Publisher.....	17
6.10.2	Insert Log Records.....	17
6.10.3	Insert Audit Records	17
6.10.4	Insert Journal Records	17
6.10.5	Check for Java Publisher Records	18
6.10.6	Start Java Publisher.....	18
6.10.7	Start Java Subscriber.....	18
6.11	Run C Publisher (jald) with GDB	18
6.12	Run C Publisher (jald) with Valgrind	18
7	JALoP Data-Taps.....	18
7.1	jalop-coreutils.....	19
7.1.1	Build & Install	19
7.1.2	Run tee and tail	19
7.2	jalop-jalauditd.....	19
7.2.1	Build & Install	19
7.2.2	Configure jalauditd.....	20
7.2.3	Run jalauditd	20
7.3	jalop-rsyslog.....	21
7.3.1	Build & Install	21
7.3.2	Configure omjal plugin	21
7.3.3	Install omjal plugin	22
7.4	jalop-log4cxx.....	23
7.4.1	Build & Install	23
7.4.2	Create, Build & Run a Test Executable	24
7.5	jalop-jaljournalld.....	25
7.5.1	Build & Install	25
7.5.2	Configure jaljournalld.....	25
7.5.3	Run jaljournalld	26
8	Seccomp (Secure Computing).....	27
8.1	JALoP Programs that are seccomp enabled and use JalSeccompEnforcer	27

8.2	JalSeccompEnforcer - api.....	27
8.3	Positioning of the JalSeccompEnforcer code in JALoP programs.....	27
8.4	Configuration Settings.....	27
8.5	How to initialize your rule sets.....	27
8.6	Test your rule sets	28
8.7	When using seccomp_debug = true on RHEL 8 and RHEL 9.....	28
8.8	When using seccomp_debug = true on RHEL 7.....	28
8.9	Using the Ansible playbook	28

This Page is Intentionally Left Blank

1 JALoPv1.x

JALoPv1.x is JALoP over BEEP. It has a C Publisher/Subscriber, a Java Publisher/Subscriber, and several data-taps. The Publisher usually resides on a CDS system to securely and reliably transfer journal, audit, and log records generated by the CDS to one or more remote Subscribers.

2 Download/Clone Source Code

Create a top-level ‘jalop’ directory to store all JALoP repositories. This will be referred to as <jalop_root> throughout this document.

```
% mkdir jalop
% cd jalop/
```

JALoP Publisher and Subscriber repositories are available on GitHub at this URL – <https://github.com/JALoP>

Clone “JALoP” and “jjnl” repositories using git –

```
% git clone https://github.com/JALoP/JALoP.git
% cd JALoP/
% git checkout -t origin/1.x.x.x

% git clone https://github.com/JALoP/jjnl.git
% cd jjnl/
% git checkout -t origin/1.x.x.x
```

The “jjnl” repository requires Java BEEP Core, which is also available on GitHub.

```
% git clone https://github.com/JALoP/beepcore-java.git
```

Initialize Git Large File storage (LFS) for the JALoP git repo

```
% sudo yum install git-lfs
% cd <jalop_root>/JALoP/
% git lfs install
% git lfs env
% git lfs pull
```

3 Build and Install C Publisher (jald) & C Subscriber (jal_subscribe)

The C Publisher (jald) is the process that negotiates with the Subscriber(s) and sends JAL records to them. This is in “jalop” (or “JALoP” if cloned from GitHub) repository. Clone the repository as mentioned above, if not done yet.

3.1 System Provisioning

3.1.1 Install Rust Dependencies

NOTE: This section can be skipped if the pre-built JALoP-rust-deps-*.rpm are present under <jalop_root>/3rd-party/rust/RHEL{7,8,9}.

JALoP uses Rust and requires the following steps to set up the required dependencies. These steps must be run on a RHEL 9 machine/VM first, with the resulting generated dependency tar.gz files copied to RHEL 7 or RHEL 8 if desired to build on these RHEL versions.

- Install EPEL
 - `$ sudo yum install epel-release # if it is not installed already.`
- Run the following script in <jalop_root>/JALoP/ to package all of the rust dependencies needed on RHEL 7/8/9.
 - `$ cd <jalop_root>/JALoP`
 - `$ sudo sh create_cargo_vendor_files.sh`
- Transfer the resulting “vendor-cargo-<rhel version>.tar.gz” file to the <jalop_root>/JALoP/ dir on the machine running the specific RHEL version of RHEL. Note: This step can be skipped on RHEL 9, as the previous step already generated the RHEL 9 version of the JALoP rust dependencies in <jalop_root>/JALoP/.
- Run the following script in <jalop_root>/JALoP/
 - `$ cd <jalop_root>/JALoP`
 - `$ sudo sh install_cargo_vendor_files.sh`

3.1.2 RHEL 7 / CentOS 7

The following instructions will allow a CentOS 7 minimal install to be provisioned to build and run the JALoP C implementation:

- Install EPEL
 - `$ sudo yum install epel-release`
- Install JALoP dependencies available from repos
 - `$ sudo yum install @development libxml2-devel libconfig-devel libuuid-devel openssl-devel xmlsec1-openssl-devel python2-scons lcov libtool-ltdl-devel doxygen libseccomp-devel systemd-devel libcap-devel libcap-ng-devel lmdb-devel boost-devel`
- Install bundled JALoP dependencies from provided RPMs
 - Navigate into your cloned JALoP repo
 - Install axl RPM
 - `$ sudo yum install 3rd-party/axl/RHEL7/*.x86_64.rpm`
 - Install vortex RPM
 - `$ sudo yum install 3rd-party/vortex/RHEL7/*.x86_64.rpm`
 - Install JALoP Rust dependency RPM (if present)
 - `$ sudo yum install 3rd-party/rust/RHEL7/*.x86_64.rpm`
- Download and install test-dept unit test library.
 - `$ git clone https://github.com/norrby/test-dept.git`
 - `$ cd test-dept`
 - `$ ./bootstrap`
 - `$ ./configure`
 - `$ sudo make install`
- NOTE: axl/vortex build steps below are only needed if these dependencies were not already installed via the provided RPMs in the steps above.
- Build and install axl
 - `$ git clone https://github.com/ASPLes/libaxl.git`
 - `$ cd libaxl`
 - `$ ./autogen.sh`
 - `$ sudo make install`
- Build and install vortex
 - `$ git clone https://github.com/ASPLes/libvortex-1.1.git`

- `$ cd libvortex-1.1`
- `$ ./autogen.sh --disable-{sasl,websocket,xml-rpc,tunnel,pull,http,alive}-support --disable-{py,lua}-vortex --disable-vortex-client`
- `$ sudo make install`

3.1.3 RHEL 8 / CentOS 8

The following instructions will allow a RHEL 8/CentOS Stream 8 minimal install to be provisioned to build and run the JALoP C implementation:

- RHEL 8: Enable CodeReady Builder
 - `$ sudo subscription-manager repos --enable codeready-builder-for-rhel-8-x86_64-rpms`
- CentOS 8: Enable Powertools (unbranded version of RHEL's CodeReady Builder)
 - `$ sudo dnf config-manager --set-enabled powertools`
- Install JALoP dependencies available from repos
 - `$ sudo dnf install @development libxml2-devel libconfig-devel loglibuuid-devel openssl-devel xmlsec1-openssl-devel libtool-ltdl-devel apr-util-devel libcurl-devel doxygen lcov libseccomp-devel systemd-devel libcap-devel libcap-ng-devel lmdb-devel boost-devel`
- Install bundled JALoP dependencies from provided RPMs
 - Navigate into your cloned JALoP repo
 - Install axl RPM
 - `$ sudo yum install 3rd-party/axl/RHEL8/*.x86_64.rpm`
 - Install vortex RPM
 - `$ sudo yum install 3rd-party/vortex/RHEL8/*.x86_64.rpm`
 - Install JALoP Rust dependency RPM (if present)
 - `$ sudo yum install 3rd-party/rust/RHEL8/*.x86_64.rpm`
- Build and install test-dept
 - `$ git clone https://github.com/norrby/test-dept.git`
 - `$ cd test-dept`
 - `$ ./bootstrap`
 - `$ ./configure`
 - `$ sudo make install`
- NOTE: axl/vortex build steps below are only needed if these dependencies were not already installed via the provided RPMs in the steps above.
- Build and install axl
 - `$ git clone https://github.com/ASPLes/libaxl.git`
 - `$ cd libaxl`
 - `$ ./autogen.sh --disable-py-axl`
 - `$ sudo make install`
- Build and install vortex
 - `$ git clone https://github.com/ASPLes/libvortex-1.1.git`
 - `$ cd libvortex-1.1`
 - `$ ./autogen.sh --disable-{sasl,websocket,xml-rpc,tunnel,pull,http,alive}-support --disable-{py,lua}-vortex --disable-vortex-client`
 - `$ sudo make install`
- Install python36-scons
 - `$ sudo yum install python36-scons`
 - `$ sudo pip3 install scons`
- Use python36 as python
 - `$ sudo alternatives --config python`

- Enter the number for the /usr/bin/python36 selection
- Ensure /usr/local/lib is in the dynamic linker path
 - `$ echo /usr/local/lib | sudo tee /etc/ld.so.conf.d/usr-local-lib.conf`
 - `$ sudo ldconfig`

3.1.4 RHEL 9

The following instructions will allow a RHEL 9 minimal install to be provisioned to build and run the JALoP C implementation:

- RHEL 9: Enable CodeReady Builder
 - `$ sudo subscription-manager repos --enable codeready-builder-for-rhel-9-x86_64-rpms`
- Install JALoP dependencies available from repos
 - `$ sudo dnf install @development libxml2-devel libconfig-devel libuuid-devel openssl-devel xmlsec1-openssl-devel libtool-ltdl-devel apr-util-devel libcurl-devel doxygen lcov libseccomp-devel systemd-devel libcap-devel libcap-ng-devel lmbd-devel boost-devel`
- Install bundled JALoP dependencies from provided RPMs
 - Navigate into your cloned JALoP repo
 - Install axl RPM
 - `$ sudo yum install 3rd-party/axl/RHEL9/*.x86_64.rpm`
 - Install vortex RPM
 - `$ sudo yum install 3rd-party/vortex/RHEL9/*.x86_64.rpm`
 - Install JALoP Rust dependency RPM (if present)
 - `$ sudo yum install 3rd-party/rust/RHEL9/*.x86_64.rpm`
- Build and install test-dept
 - `$ git clone https://github.com/norrby/test-dept.git`
 - `$ cd test-dept`
 - `$ ./bootstrap`
 - `$ ./configure`
 - `$ sudo make install`
- NOTE: axl/vortex build steps below are only needed if these dependencies were not already installed via the provided RPMs in the step above.
- Build and install axl
 - `$ git clone https://github.com/ASPLes/libaxl.git`
 - `$ cd libaxl`
 - `$ ./autogen.sh --disable-py-axl`
 - `$ sudo make install`
- Build and install vortex
 - `$ git clone https://github.com/ASPLes/libvortex-1.1.git`
 - `$ cd libvortex-1.1`
 - `$ ./autogen.sh --disable-{sasl,websocket,xml-rpc,tunnel,pull,http,alive}-support --disable-{py,lua}-vortex --disable-vortex-client`
 - `$ sudo make install`
- Install python3-scons
 - `$ sudo yum install python3-scons`
- Ensure /usr/local/lib is in the dynamic linker path
 - `$ echo /usr/local/lib | sudo tee /etc/ld.so.conf.d/usr-local-lib.conf`
 - `$ sudo ldconfig`

3.2 Build and Install C Publisher & C Subscriber

The “jalop” (or “JALoP” if cloned from GitHub) repository builds the following applications –

- “jald” (the JALoP C Publisher)
- “jal-local-store” (the JAL-Local-Store)
- “jaldb_tail”
- “jal_dump”
- “jal_purge”
- “jalp_test” (a test Producer)
- “jal_subscribe” (the JALoP C Subscriber)
- “jaldb_inline_filter” (the JALoP Rust Inline Filter)

It also builds the following shared libraries –

- “libjal-common.so”
- “libjal-db.so”
- “libjal-network.so”
- “libjal-producer.so”
- “libjal-utils.so”

Follow the instructions below to build and install the above-mentioned components -

- Change to the jalop top directory –
 - `% cd <jalop_root>/jalop/` (or, `cd <jalop_root>/JALoP/`, if cloned from github).
- Checkout the “1.x.x.x” branch of jalop (if not already checked out. Check “git branch” output) -
 - `% git checkout 1.x.x.x`
- Clean up jalop -
 - `% scons -c` # to clean up.
- Build jalop and install publisher and subscribers using the LMDB database
 - `% scons`
 - `sudo ./install_rhel_x86_64.sh`

3.3 Configure C Publisher

Make a copy of `<jalop_root>/jalop/test-input/jald.cfg` and adjust according to your Subscriber’s IP, port, etc. This is to avoid changing the sample configuration file in the git source tree.

Check the section “Run the Publisher and Subscriber to Transfer JAL Records” below.

3.4 Configure JAL-Local-Store

The JAL-Local-Store (`jal-local-store`) is a process that receives and stores JAL Data sent from the JAL Producer applications. It has a Lightning Memory-Mapped Database (LMDB) to store and process the JAL records. The `jal-local-store` process must be started before the Publisher process (`jald`).

Make a copy of `<jalop_root>/jalop/test-input/local_store.cfg` and update accordingly. This is to avoid changing the sample configuration file in the git source tree.

3.5 Configure C Subscriber

Make a copy of `<jalop_root>/jalop/test-input/jal_subscribe.cfg` and update accordingly. This is to avoid changing the sample configuration file in the git source tree. Configure LMDB (Optional)

Optionally, a file named `LMDB_CONFIG` can be placed at the `<db_root>` directory to override LMDB specific configuration settings. The two configuration settings that can be overridden are `database_option` and `lmdb_map_size`. If the `LMDB_CONFIG` file is not present and/or does not contain these settings, then default values are used. For `lmdb_map_size`, which sets the maximum allowed size of the `lmdb` database, the default is 120 gigabytes. For `database_option`, the default is LMDB performance level 2 (`JDB_LMDB_PERFORMANCE_LEVEL2`). These settings are global across all JAL applications.

Make a copy of `<jalop_root>/jalop/test-input/LMDB_CONFIG` into the `db` root directory (which is the directory specified by the `db_root` config setting in the local store configuration file) and update accordingly.

4 Build and Install BeepCore Java

4.1 System Provisioning

BeepCore Java is a Java implementation of BEEP (RFC3080 and RFC3081) and is required by JJNL.

Install the following packages (if not done already) –

- Java 8 or newer
- Ant
- Maven

```
% sudo yum install java-1.8.0-openjdk-devel ant maven
```

4.2 Clone “beepcore-java” repository from GitHub

```
% cd <jalop_root>/
% git clone https://github.com/JALoP/beepcore-java.git
```

Build and Install BeepCore JavaBuild –

```
% cd <jalop_root>/beepcore-java/
% ant dist-tgz
```

Install –

```
% cd build/beepcore-0.9.20/lib/
% ./install_jars.sh
```

This should install 3 JAR files to `~/m2/repository/` as shown in the example output below –

```
[INFO] Installing /home/marefin/jalop/beepcore-java/beepcore.jar to
/home/marefin/.m2/repository/org/beepcore-
java/beepcore/0.9.20/beepcore-0.9.20.jar
```

```
[INFO] Installing /home/marefin/jalop/beepcore-java/bbeptls-jsse.jar
to /home/marefin/.m2/repository/org/beepcore-java/bbeptls-
jsse/0.9.20/bbeptls-jsse-0.9.20.jar
```

```
[INFO] Installing /home/marefin/jalop/beepcore-java/concurrent.jar to
/home/marefin/.m2/repository/EDU/oswego/cs/dl/util/concurrent/1.3.4/co
ncurrent-1.3.4.jar
```

Verify that the 3 JAR files mentioned above are successfully installed; if for some reason those are not installed, manually install (copy) them to intended destination as shown above.

4.3 Install BeepCore Java from pre-built jar files

Extract –

```
% cd <jalop_root>/jjnl/3rd-Party/beepcore-java/0.9.20
% tar -xvzf beepcore-0.9.20.tgz
```

Install –

```
% cd ./beepcore-0.9.20/lib/
% sh ./install_jars.sh
```

This should install 3 JAR files to ~/.m2/repository/ as shown in the example output below –

```
[INFO] Installing /home/marefin/jalop/beepcore-java/beepcore.jar to
/home/marefin/.m2/repository/org/beepcore-
java/beepcore/0.9.20/beepcore-0.9.20.jar
```

```
[INFO] Installing /home/marefin/jalop/beepcore-java/bbeptls-jsse.jar
to /home/marefin/.m2/repository/org/beepcore-java/bbeptls-
jsse/0.9.20/bbeptls-jsse-0.9.20.jar
```

```
[INFO] Installing /home/marefin/jalop/beepcore-java/concurrent.jar to
/home/marefin/.m2/repository/EDU/oswego/cs/dl/util/concurrent/1.3.4/co
ncurrent-1.3.4.jar
```

Verify that the 3 JAR files mentioned above are successfully installed; if for some reason those are not installed, manually install (copy) them to intended destination as shown above.

5 Build and Install Java Subscriber and Publisher (jnl_test-*.jar)

5.1 Clone “jjnl” repository from GitHub

This is the “jjnl” repository. Clone the repository as below (if not done already) –

```
% cd <jalop_root>/
```

```
% git clone https://github.com/JALoP/jjnl.git
% cd jjnl
% git checkout -t origin/1.x.x.x
```

5.2 Build and Install Java Subscriber (jnl_test)

- Build -
 - % cd <jalop_root>/jjnl/jnl_parent/
 - % mvn clean
 - % mvn package
- Install (optional) -
 - % cd <jalop_root>/jjnl/jnl_lib/
 - % mvn install

Note that “jnl_test-*.jar” can be configured and run either as a Subscriber or as a Publisher, as described in the subsequent sections.

5.3 Configure jnl_test-*.jar as a Subscriber

```
% cd <jalop_root>/jjnl/jnl_test/
```

Copy ./src/test/resources/sampleSubscriber.json here and update accordingly. This is to avoid changing the sample config file in git source tree. If not running with TLS, comment out or remove the “ssl” section below.

Check the section “Run the Publisher and Subscriber to Transfer JAL Records” below.

5.4 Configure jnl_test-*.jar as a Publisher

Known Limitation: The jnl_test-*.jar Java Publisher does not support batching of digest-challenge. Therefore, set the batch size (pendingDigestMax) to 1.

6 Run Publisher and Subscriber to Transfer JAL Records

Follow the steps below to run the C Publisher and Java Subscriber to transfer JAL records.

6.1 Disable SELinux or set to Permissive (For Testing Only)

```
=====
Permanent: have "SELINUX=disabled" or "SELINUX=permissive" in
"/etc/selinux/config" file, then restart VM.
Temporary: Enter the command "/usr/sbin/setenforce 0"
```

6.2 Stop Firewall (For Testing Only)

```
=====
RHEL/CentOS 6.x
    Stop:      % sudo service iptables stop
    Disable:   % sudo chkconfig iptables off
```

```
RHEL/CentOS 7.x, 8.x, 9.x
    Stop:      % sudo service firewalld stop
    Or,        % sudo systemctl stop firewalld
    Disable:   % sudo systemctl disable firewalld
```

6.3 Start JAL-LOCAL-STORE

```
=====
# May need to clean up the first time. Make sure no jal-local-store is
running. For examples -
$ cd <jalop_root>/jalop/
$ pkill jal-local-store
$ sudo rm -rf ./jal.sock      # remove old JAL socket, if any.
$ sudo rm -rf /root/testdb   # clean up existing <db_root>/ directory.
$ sudo mkdir /root/testdb    # For the first time, create <db_root>/
```

Make a copy of ./test-input/local_store.cfg here and update accordingly. This is to avoid changing the sample configuration file in the git source tree.

```
$ sudo ./release/bin/jal-local-store --debug --no-daemon -c
./local_store.cfg &
```

6.4 Insert Records into JAL-Local-Store

```
$ cd <jalop_root>/jalop/
```

6.4.1 Insert Log Records

```
=====
$ sudo ./release/bin/jalp_test -j ./jal.sock -a ~/jalop/jalop/test-
input/sample2.cfg -p ~/jalop/jalop/test-input/big_payload.txt -n 100 -
t 1
```

6.4.2 Insert Audit Records

```
=====
$ sudo ./release/bin/jalptest -j ./jal.sock -a ~/jalop/jalop/test-
input/sample2.cfg -p ~/jalop/jalop/test-input/big_payload.txt -n 100 -
t a
```

6.4.3 Insert Journal Records

```
=====
$ sudo ./release/bin/jalptest -j ./jal.sock -a ~/jalop/jalop/test-
input/sample2.cfg -p ~/jalop/jalop/test-input/big_payload.txt -n 100 -
t j
```

Note: Use "-t f" instead of "-t j" if you want JAL Producer to send journal file descriptor through the Unix Domain Socket (instead of the payload bytes) to JAL-Local-Store. Passing file descriptor may perform better for larger journal payloads (e.g., in the order of megabytes).

6.5 Check for Inserted Records in JAL-Local-Store

```
=====
$ sudo ./release/bin/jaldb_tail -n 100 -h /root/testdb/ -t l | wc -l
$ sudo ./release/bin/jaldb_tail -n 100 -h /root/testdb/ -t a | wc -l
$ sudo ./release/bin/jaldb_tail -n 100 -h /root/testdb/ -t j | wc -l
```

Or,

```
$ sudo ./release/bin/jalpurge -h /root/testdb -b 2030-11-11T11:11:11
-x -t l | wc -l
$ sudo ./release/bin/jalpurge -h /root/testdb -b 2030-11-11T11:11:11
-x -t a | wc -l
$ sudo ./release/bin/jalpurge -h /root/testdb -b 2030-11-11T11:11:11
-x -t j | wc -l
```

6.6 Start Rust Inline Filter (Recommended)

```
=====
$ cd <jalop_root>/jalop/
```

NOTE: If verbose logging is desired optionally run the following command below.

```
$ export RUST_LOG=trace
```

```
$ sudo rm -f jald_filter_socket_*
```

```
$ sudo ./release/bin/jaldb_inline_filter -c test-input/jaldb-inline-
filter.toml
```

Note: If digest batching is being used by any subscriber (pending_digest_max greater than 1 in any subscriber config file), and slow record transfer is experienced when using the jaldb_inline_filter, then the record-socket "buffer" config setting

for the inline filter can be increased. The default is 256 which should be adequate for normal use cases. This must be equal or greater than the total of all the pending_digest_max settings of all connected subscribers.

Ex: pending_digest_max is set to 100 for three subscribers, then -socket-buffer-size must be set to 300 or higher. If slow record transfer is experienced, this number can be increased further.

Enter `“./release/bin/jaldb_inline_filter -h”` for help.

6.7 Start C Publisher

```
=====
$ cd <jalop_root>/jalop/
```

Make a copy of `./test-input/jald.cfg` here and update accordingly. This is to avoid changing the sample configuration file in the git source tree.

```
$ sudo ./release/bin/jald -f -d -c ./jald.cfg --no-daemon 2>&1 | tee publisher.log
```

NOTE: To run jald without using the inline filter use the following command below instead (no -f option). It is not recommended to run jald publisher without the inline filter.

```
$ sudo ./release/bin/jald -d -c ./jald.cfg --no-daemon 2>&1 | tee publisher.log
```

Enter `“man jald”` for help.

6.8 Start Subscriber

```
=====
Java subscriber (jnl_test)
=====
$ cd <jalop_root>/jjnl/jnl_test/
```

Make a copy of `./jnl_test/target/test-classes/sampleHttpSubscriber.json` here and update accordingly. This is to avoid changing the sample configuration file in the git source tree.

```
java -jar target/jnl_test-1.1.0.1.jar ./sampleSubscriber.json 2>&1 | tee subscriber.log
```

```
=====
or C subscriber (jal_subscribe)
=====
$ cd <jalop_root>/jalop
```

Make a copy of ./test-input/jal_subscribe.cfg here and update accordingly. This is to avoid changing the sample configuration file in the git source tree. If running on same machine as publisher, you need to specify a different location for the database. The publisher and subscriber need to use separate databases.

```
sudo ./release/bin/jal_subscribe -d -c ./jal_subscribe.cfg 2>&1 | tee subscriber.log
```

Enter "man jal_subscribe" for help.

6.9 Purge Records from Local Store

```
=====
$ sudo ./release/bin/jal_purge -h ./testdb/ -d -f -c -b 2030-11-11T11:11:11 -t l
$ sudo ./release/bin/jal_purge -h ./testdb/ -d -f -c -b 2030-11-11T11:11:11 -t a
$ sudo ./release/bin/jal_purge -h ./testdb/ -d -f -c -b 2030-11-11T11:11:11 -t j
```

OR,

```
$ now=$(date -u "+%Y-%m-%dT%H:%M:%S.%N")
$ sudo ./release/bin/jal_purge -h ./testdb/ -d -f -c -b "$now" -t l
$ sudo ./release/bin/jal_purge -h ./testdb/ -d -f -c -b "$now" -t a
$ sudo ./release/bin/jal_purge -h ./testdb/ -d -f -c -b "$now" -t j
```

6.10 Run Java Publisher and Java Subscriber

```
=====
```

Follow the steps below to run the Java Publisher and Java Subscriber to transfer JAL records.

6.10.1 Add Records to Java Publisher.

```
$ cd <jjnl_root>/
$ mkdir input
```

6.10.2 Insert Log Records

```
=====
$ python generate_records.py log input/ 100 ~/jalop/jalop/test-input/sample2.cfg ~/jalop/jalop/test-input/big_payload.txt
```

6.10.3 Insert Audit Records

```
=====
$ python generate_records.py audit input/ 100 ~/jalop/jalop/test-input/sample2.cfg ~/jalop/jalop/test-input/big_payload.txt
```

6.10.4 Insert Journal Records

```
=====
```

```
$ python generate_records.py journal input/ 100 ~/jalop/jalop/test-  
input/sample2.cfg ~/jalop/jalop/test-input/big_payload.txt
```

6.10.5 Check for Java Publisher Records

```
=====
$ ls <jjnl_root>/input/audit
$ ls <jjnl_root>/input/journal
$ ls <jjnl_root>/input/log
```

6.10.6 Start Java Publisher

```
=====
$ cd <jjnl_root>/jnl_test/
```

Make a copy of `./src/test/resources/sampleListener.json` here and update accordingly. This is to avoid changing the sample configuration file in the git source tree.

```
$ java -jar jnl_test-1.1.01.jar sampleListener.json
```

6.10.7 Start Java Subscriber

```
=====
$ cd <jalop_root>/jjnl/jnl_test/
```

Make a copy of `./jnl_test/target/test-classes/sampleSubscriber.json` here and update accordingly. This is to avoid changing the sample configuration file in the git source tree.

```
java -jar target/jnl_test-1.1.0.1.jar ./sampleSubscriber.json 2>&1 |  
tee subscriber.log
```

6.11 Run C Publisher (jald) with GDB

```
=====
$ export LD_LIBRARY_PATH="/home/marefin/jalop/jalop/debug/lib/"
$ gdb -ex=r --args ./debug/bin/jald -d -c ./jald.cfg --no-daemon
```

6.12 Run C Publisher (jald) with Valgrind

```
=====
$ valgrind --tool=memcheck --leak-check=full --verbose --track-  
origins=yes --log-file=valgrind_out.txt ./debug/bin/jald -d -c  
./jald.cfg --no-daemon
```

7 JALoP Data-Taps

The JALoP data-taps capture and send JAL records to the JALoP Local Store (`jal-local-store`). Each data-tap depends on JALoP libraries, so JALoP v1 or v2 libraries must be build and installed prior to building a data-tap repository. Generally, each of the data-taps requires the JALoP socket and `db_root` addresses that `jal-local-store` (JLS) uses. Note that `jal-local-store` must be already running for any data-tap to send JAL records to it via the socket.

7.1 jalop-coreutils

Available on IntelLink and JALoP development Git Server.

This has the JALoP version of GNU tail and GNU tee commands.

7.1.1 Build & Install

```
% cd <jalop_root>/jalop-coreutils/  
% autoreconf -fiv #for RHEL7  
% ./bootstrap --skip-po #for RHEL8/RHEL9  
% ./configure --disable-gcc-warnings  
% make -j  
% make check  
% [[ -e /tmp/install ]] || mkdir /tmp/install  
% make install DESTDIR=/tmp/install  
% cp /tmp/install/usr/local/bin/{tee,tail} /usr/local/bin
```

7.1.2 Run tee and tail

Note that the jal-local-store (JLS) process must be running already to receive and insert records.

```
# Based on <db_root> location, user may need 'sudo' in the  
following instructions.  
  
% echo "This is a test message to tee into JLS" | tee -j --  
path=<path to JALoP socket> --schema-root=<path to JALoP schema  
directory>  
% echo -e "Test message1\nTest message2" > file_to_tail  
$ tail -j --path=<path to JALoP socket> --schema-root=<path to  
JALoP schema directory> ./file_to_tail
```

Use `jaldb_tail` to check for newly inserted records -

```
% jaldb_tail -t 1 -h <db_root>
```

Use `jal_dump` to verify that the log records came from auditd -

```
% jal_dump -u <UUID found using jaldb_tail> -h <db_root> -t 1 -d  
z
```

7.2 jalop-jalauditd

Available on the NCDSMO IntelLink site and JALoP development Git Server. This is also available at [GitHub.com/JALoP/JALoP-Auditd-Plugin](https://github.com/JALoP/JALoP-Auditd-Plugin).

7.2.1 Build & Install

Note that JALoP v1 or v2 must be built and installed before attempting to build this data tap.

```
% cd <jalop_root>/jalop-jalauditd (or JALoP-Auditd-Plugin)  
% make clean  
% make  
% sudo make install
```

This shall install the followings –

- The binary `/sbin/jalauditd` (or `/usr/sbin/jalauditd`)
- The configuration files for RHEL7 and RHEL8/RHEL9 go into different directories. Make install will install the configuration files in the following locations and will need to be copied to appropriate directory:
 - `/etc/jalauditd/jalauditd.conf`
 - `/etc/audit/plugins.d/audisp-jalauditd.conf`
- For RHEL7 the configuration files should be:
 - `/etc/audisp/jalauditd.conf`
 - `/etc/audisp/plugins.d/audisp-jalauditd.conf`
- For RHEL8/RHEL9 the configuration files should be:
 - `/etc/jalauditd/jalauditd.conf`
 - `/etc/audit/plugins.d/audisp-jalauditd.conf`
- The `jalauditd.conf` file is initially empty; you will need to edit this file, see below.

7.2.2 Configure jalauditd

The configuration file `jalauditd.conf` is initially empty. There can be 4 settings in this file as shown below.

```
socket = "/path/to/jalop/socket";
schemas = "/path/to/schemas/root";
keypath = "/path/to/key";
certpath = "/path/to/cert";
```

If the `socket` and `schemas` locations are not specified above, default locations specified by the JAL Producer Library (JPL) will be used. Below are the default socket and schemas locations –

```
socket = "/var/run/jalop/jalop.sock"
schemas = "/usr/share/jalop/schemas"
```

If `keypath` or `certpath` are not specified, no key or cert will be used for signing.

IMPORTANT: These settings must be consistent with the `jal-local-store` configuration file.

7.2.3 Run jalauditd

Note that the `jal-local-store` (JLS) process must run to receive and insert records sent by `jalauditd` via the socket.

Restarting (as root or sudo) `auditd` automatically runs the `jalauditd` child process.

```
% service auditd restart (or, systemctl restart auditd)
```

Use `jaldb_tail` to check for newly inserted records -

```
% sudo jaldb_tail -f -t 1 -h <db_root>
```

Use `jal_dump` to verify that the log records came from `auditd` -

```
% sudo jal_dump -u <UUID found using jaldb_tail> -h <db_root> -t 1  
-d z
```

See <jalop_root>/jalop-jalauditd/README file for more details.

7.3 jalop-rsyslog

Available on IntelLink and JALoP development Git Server.

7.3.1 Build & Install

Note that JALoP must be built and installed before attempting to build the rsyslog plugin.

```
$ sudo yum install git automake libtool libestr-devel
```

If you have access to a libfastjson-devel package, you can install that and skip to **Error! Reference source not found.**:

```
$ sudo yum install libfastjson-devel
```

7.3.1.1 Libfastjson headers

If you cannot install libfastjson-devel, the libfastjson headers must be made available to the rsyslog plugin in another way.

```
$ git clone https://github.com/rsyslog/libfastjson.git  
$ cd libfastjson  
$ sudo mkdir /usr/include/libfastjson  
$ sudo cp ./*.h /usr/include/libfastjson  
$ sudo ln -s /usr/lib64/libfastjson.so.4  
/usr/lib64/libfastjson.so
```

On RHEL 7 / CentOS 7:

```
$ export JSON_C_FLAGS="-g"  
$ export JSON_C_LIBS="/usr/lib64/libfastjson.so"
```

On RHEL 8 / CentOS Stream 8:

```
$ export LIBFASTJSON_CFLAGS="-g"  
$ export LIBFASTJSON_LIBS="/usr/lib64/libfastjson.so"
```

7.3.1.2 Build omjal plugin

Build the omjal plugin:

```
$ cd jalop-rsyslogd  
$ export CPPFLAGS="-I/usr/include/libfastjson"  
$ sh autogen.sh  
$ make
```

7.3.2 Configure omjal plugin

Add the following configuration section to /etc/rsyslog.conf:

```

$ModLoad omjal # calls JAL for rsyslog pass through to JALoP
$ActionOmjalSocket /var/run/jalop/jalop.sock
$ActionOmjalSchemas /usr/share/jalop/schemas
$ActionOmjalKey <path_to_key> # Optional
$ActionOmjalCert <path_to_key> # Optional
*. * :omjal:

```

\$ModLoad omjal

States that the omjal plugin shall be loaded

\$ActionOmjalSocket /var/run/jalop/jalop.sock

Specifies the socket to be used to forward the messages to the JALoP local store. If this option is omitted from the file, it will use the default value specified in the JALoP library.

\$ActionOmjalSchemas /usr/share/jalop/schemas

Specifies the directory where the schema information is located. If this option is omitted from the file, it will use the default value specified in the JALoP library.

\$ActionOmjalKey <path_to_key>

Specifies the path to the key to be used for signing. If this option is omitted from the file, then no key will be used.

\$ActionOmjalCert <path_to_key>

Specifies the path to the certificate to be used for signing. If this option is omitted from the file, then no certificate will be used.

***. * :omjal:logJAL**

<msg_filter_expression>:<output_module>:<parameters>;<template_name>

*. * specifies that all messages will be sent to this module. If, for example, you wanted to handle mail messages, mail.* would handle that.

See <https://www.rsyslog.com/doc/v8-stable/configuration/filters.html> for more information.

7.3.3 Install omjal plugin

Install the omjal plugin:

```

$ cd jalop-rsyslogd
$ make install

```

The rsyslog package installs libraries to /usr/lib64/rsyslog. If your environment uses a different location, either manually copy the library from .libs/omjal.so or change the libdir flag in autogen.sh

After installation, and after starting the **jal-local-store**, restart the rsyslog service:

```

$ sudo systemctl restart rsyslog

```

To ensure that rsyslogd is connecting to jal-local-store, you can run jal-local-store with the debug (-d) flag. When jal-local-store is ready, the message `Thread_count: 1` will appear. After rsyslogd is connected, the message `Thread_count: 2` will appear.

To create messages in rsyslogd, the logger command can be used. For example:

```
logger "This is your message content."
```

Use `jaldb_tail` to check for newly inserted records -

```
% sudo jaldb_tail -f -t 1 -h <db_root>
```

Use `jal_dump` to verify that the log records came from auditd -

```
% sudo jal_dump -u <UUID found using jaldb_tail> -h <db_root> -t  
1 -d z
```

7.4 jalop-log4cxx

Available on IntelLink and JALoP development Git Server.

7.4.1 Build & Install

Note that JALoP must be built and installed before attempting to build the jalop-log4cxx library.

```
$ sudo yum install apr-util-devel
```

On RHEL 7 / CentOS 7:

```
$ sudo yum install cmake3  
$ sudo yum install centos-release-scl  
$ sudo yum install devtoolset-8-gcc-c++ --  
enablerepo='centos-scl-rh'  
$ sudo scl enable devtoolset-8 'bash'
```

On RHEL 8 / CentOS 8/ RHEL 9:

```
$ sudo yum install cmake
```

Build the library:

On RHEL 7 / CentOS 7:

```
$ cd jalop-log4cxx  
$ cmake3 -D CMAKE_CXX_COMPILER=/opt/rh/devtoolset-  
8/root/usr/bin/g++ -D CMAKE_INSTALL_PREFIX=/usr CMakeLists.txt  
$ make
```

On RHEL 8 / CentOS 8 / RHEL 9:

```
$ cd jalop-log4cxx  
$ cmake -D CMAKE_INSTALL_PREFIX=/usr CMakeLists.txt  
$ make
```

Install the library:

```
$ sudo make install
```

To uninstall the library:

```
$ sudo make uninstall
```

7.4.2 Create, Build & Run a Test Executable

An example test executable is created during the CMake build at `/path/to/jalop-log4cxx/src/examples/cpp/jalp-appender`. To test using this executable, skip ahead to the instructions on starting the `jal_local_store` and running the executable.

To create a new test executable, do the following:

Create a separate folder for the test executable. This can be separate from the `jalop-log4cxx` folder:

```
$ mkdir ~/log4cxx-test  
$ cd ~/log4cxx-test
```

Copy required files from `jalop-log4cxx`:

```
$ cp /path/to/jalop-log4cxx/src/main/cpp/jalpappender.cpp .
```

Create a `main.cpp` for the test executable. Use the following as an example:

```
#include <log4cxx/logger.h>  
#include <log4cxx/propertyconfigurator.h>  
#include <log4cxx/jalpappender.h>  
  
using namespace log4cxx;  
  
LoggerPtr logger(Logger::getLogger("Test"));  
  
int main(int argc, char **argv)  
{  
    PropertyConfigurator::configure(argv[1]);  
    LOG4CXX_INFO(logger, "qwerty")  
    return 0;  
}
```

Build the test executable:

```
$ g++ main.cpp jalpappender.cpp -llog4cxx -ljaval-producer -o  
log4cxx-test
```

Create a configuration file for the test executable. Use the following as a starting point:

```
log4j.rootLogger=DEBUG,A1  
  
log4j.appender.A1=org.apache.log4j.JalpAppender
```

```
log4j.appender.A1.layout=org.apache.log4j.PatternLayout
log4j.appender.A1.layout.ConversionPattern=%-4r [%t] %-5p %c %x -
%m%n
```

The following are optional parameters than can be added to the log4cxx config file:

After starting the **jal-local-store**, run the test executable:

```
$ ./log4cxx-test path/to/test.cfg
```

This will send one record to jal-local-store before closing the connection. Verify the record with the following:

```
$ jaldb_tail -h /path/to/testdb -t 1 -d a
```

```
log4j.appender.A1.PATH - The path to jalop.sock
log4j.appender.A1.HOSTNAME - The hostname of the machine that the
JALoP local store is on
log4j.appender.A1.APPNAME - The name of the process that is logging
data
log4j.appender.A1.SCHEMA_ROOT - The path to the JALoP schemas
log4j.appender.A1.KEY_PATH - The path to the RSA key used for signing
log4j.appender.A1.CERT_PATH - The path to the Certificate used for
signing
```

This will print the app metadata for the records. The record inserted by the test executable does not create a payload from the string in LOG4CXX_INFO. It ends up in the message tag in the app metadata.

7.5 jalop-jaljournald

Available on NCDSMO IntelLink site and JALoP development GitLab Server.

7.5.1 Build & Install

Dependencies: JALoP v1 or v2 shared libraries must be built and installed.

- /usr/lib64/<jalop shared object files>
- /usr/include/<jalop headers>
- systemd-devel rpm

```
% cd <jalop_root>/jalop-jaljournald
% make clean
% make
% sudo make install
```

This shall install the followings –

- The binary /bin/jaljournald

- The `jaljournalld` configuration file `/etc/jalop/jaljournalld.cfg`. This file initially has some default values; you will need to edit this file, see below.

7.5.2 Configure `jaljournalld`

The `jalauditd` configuration file `/etc/jalauditd/jalauditd.conf` initially has some default values; The settings in this file as shown below.

```
# verbose logging
debug = false;

# Start collecting logs from head, boot, or tail. Default: tail
# head: will get all journal logs available from multiple boots
#       and then continuously poll to get all logs as they arrive
# boot: will get all logs from current boot and then continuously
#       poll to get all logs as they arrive.
# tail: will continuously poll to get all logs as they arrive.
init_point = "tail";

# collect all journal fields. Default: MESSAGE only
metadata = false;

# socket path that jal-local-store will be listening.
socket = "/path/to/jalop/socket";

# jalop schema path
schemas = "/path/to/schemas/root";

# hostname to record in log
hostname = "localhost";

# appname to record in log
appname = "jalop-journalld";

# poll delay between checking for new journal logs
# in milliseconds 1000 = 1 second
delay = 1000;
```

If the `socket` and `schema_path` locations are not specified above, default locations specified by the JAL Producer Library (JPL) will be used. Below are the default `socket` and `schema_path` locations –

```
socket = "/var/run/jalop/jalop.sock"
schema_path = "/usr/share/jalop/schemas"
```

IMPORTANT: These settings must be consistent with the `jal-local-store` configuration file.

7.5.3 Run `jaljournalld`

Note that the `jal-local-store` (JLS) process must run to receive and insert records sent by `jaljournalld` via the `socket`. Also, make sure the `jal-local-store` `socket` path matches the `jaljournalld` `socket` path and you have write permissions on that `socket` path.

Example command-line

```
% jaljournald -c <path-to-config> --debug --metadata
```

Use `jalldb_tail` to check for newly inserted records -

```
% sudo jalldb_tail -f -t 1 -h <db_root>
```

Use `jal_dump` to verify that the log records came from `journald` -

```
% sudo jal_dump -u <UUID found using jalldb_tail> -h <db_root> -t 1  
-d z
```

8 Seccomp (Secure Computing)

Seccomp is a security feature in the Linux kernel that allows a process to transition into a secure state where it can only make a limited set of system calls. This feature is useful for sandboxing applications, limiting their ability to interact with the system and reducing the attack surface.

8.1 JALoP Programs that are seccomp enabled and use JalSeccompEnforcer

- `jal-localstore`
- `jald`
- `jal_subscribe`

8.2 JalSeccompEnforcer - api

- 1) `#include "jal_seccomp_enforcer.h"`
- 2) `struct jal_seccomp_enforcer_t* seccomp_enforcer = jal_seccomp_enforcer_create(config_path);`
- 3) `jal_seccomp_enforcer_apply_initial(seccomp_enforcer);`
- 4) `jal_seccomp_enforcer_apply_final(seccomp_enforcer);`

8.3 Positioning of the JalSeccompEnforcer code in JALoP programs

- 1) A JALoP program will use the `#include`.
- 2) Calls `jal_seccomp_enforcer_create()`, right after knowing the location of the config file.
- 3) Applies initial, right after creation of the `seccomp_enforcer`.
- 4) Applies final, right after the program has finished setting up and right before the program starts to do its routine work.

8.4 Configuration Settings

The following configuration settings are available in an applicable JALoP program configuration file under the `seccomp` setting.

- `enable = true;`
- `debug = false;`
- `initial = ["arch_prctl", "bind", ...];`
- `both = ["brk", "ftruncate", ...];`
- `final = ["accept", "access", ...];`

All programs listed above have default seccomp configurations. See various sections above for complete rulesets used sample configuration files.

8.5 How to initialize your rule sets

Do the following in the applicable configuration file under the seccomp setting.

- Set enable = true;
- Set debug = true;
- Comment out any rulesets.
- Run your process with strace and try to have it exercise all its functionality
 - `strace -f -o strace.out <process_name> <process_option>`
- Run the `collect_system_calls.py` script in `<jalop_home>/src/test_utils/scripts/`
 - `collect_system_calls.py strace.out > seccomp.cfg`
- The script above will produce your rule sets.
 - `initial = ["arch_prctl", "bind", ...];`
 - `both = ["brk", "ftruncate", ...];`
 - `final = ["accept", "access", ...];`

8.6 Test your rule sets

- Run your program with `enable_seccomp = true;` and `seccomp_debug = false;`
- If the program runs to completion, that is great.
- If the process stops, the syscall that it failed on will be logged to stdout.
- Add that syscall to the initial set first and run a test again.
- If it fails again for the same syscall, move the syscall to your final set.
- If it fails again for the same syscall, move the syscall to your both set.

8.7 When using `seccomp_debug = true` on RHEL 8 and RHEL 9

- You need to also have `enable_seccomp = true;`
- There needs to be at least one rule in each set.
- Your process will run freely. Seccomp will not be enforced.
- There will be a log entry in `/var/log/audit/audit.log` for each syscall hit that is not in your rule sets.
- You can `grep` for `SYSCALL` in `/var/log/audit/audit.log`.
- The process will also create a marker syscall while working. This mark is produced at the time `jal_seccomp_enforcer_apply_final(seccomp_enforcer)` is called. The syscall can be identified in the logs by looking for "Applying final ruleset DONE".
- The mark is used by `collect_system_calls.py` to determine what calls should be placed in your initial, both and final rulesets.
- Stderr will log any syscalls in rule sets that are not valid.

8.8 When using `seccomp_debug = true` on RHEL 7

- It will do as above except
 - There is not a `SCMP_ACT_LOG` action in the libseccomp library, so no syscalls will be logged to `/var/log/audit/audit.log`.
- You need to run strace. (See above in section 8.5 How to initial your rulesets.)

8.9 Using the Ansible playbook

- There is an ansible playbook in <jalop_home>/src/test_utils/ansible/seccomp.yml.
- The playbook should already be configured properly as is, but inspect seccomp.yml and make any adjustments if needed.
- The sample configuration files used in the playbook should already be configured properly, but inspect your configuration files used and make any adjustments if needed.
- The playbook will run a thorough test of the JALoP processes.
- The playbook will also build your seccomp configurations.
- All output is produced in the <jalop_home>/test_logs directory.
- `ansible-playbook <jalop_home>/src/test_utils/ansible/seccomp.yml -e cwd=`pwd` -e mode=release -e perf=p2 -e rust_log=release -e marker=rust -e rcnt=5000 -e types=jalf`